

Ashish Tondwalkar

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EDUCATION

Rochester Institute of Technology

MS & BS Electrical Engineering

Expected 2024

MS GPA 3.88, BS GPA 3.83

Relevant Graduate Coursework: Complex Digital Systems Verification, Advanced Topics in Computer Systems Design, Design of Computer Systems, Design of Digital Systems, Real Time Embedded Systems

EXPERIENCE

Digital Design Engineering Intern @ Allegro Microsystems

May – August 2023

- Communicated with Systems and Analog teams to formulate duties of the IP's digital system. Documented specifications and macro-architecture for DV modeling. Implemented algorithms and digital design in SystemVerilog. Generalized IP for portability. Resolved inconsistencies found by DV team.

Systems Engineering Co-op @ Xilinx (now AMD)

January – August 2022

- Created Microblaze and Versal FPGA systems to run diagnostics on DDR5 memory controllers. Developed bare metal C applications to find and reproduce faulty behavior. Automated system bring-up and testing with Tcl scripting. Created behavioral models and reformatted pre-release RTL for size estimates.

Design of Computer and Digital Systems Grader @ RIT

August – December 2023

- Graded grad and undergrad HDL and hardware architecture courses. Provided feedback and criticism on designs.

Digital Systems 2 TA @ RIT

August – December 2022

- Conducted Verilog and VHDL lab sessions. Taught students how to make architectural design choices, and guided them through the debugging process.
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SELECTED PROJECTS

Heterogenous Computer System with AI Accelerator

Spring 2023

- Created a tensor subsystem with the GPP core described in the next project. The system uses DMA transfers between data memory (DM) and a FIFO architecture, which propagates data to a deterministic systolic array. The GPP D-Cache snoops writes from the tensor subsystem to the DM. A custom assembler for the system features multi-module linking, high level programming language pragma translations, and interrupt setups.

Cached Multicore Pipelined Processor and Assembler

Fall 2022

- Created a master-slave multicore system with Verilog. 4-stage pipelined cores communicate with a custom handshake protocol. Hazards are resolved with data-forwarding. Data and Program memory are implemented with 2-way Cache. Test programs are converted to machine code through a custom assembler.

Tensor Processors

Summers 2021 & 2022

- 2.0:** Developed and verified a parameterized int 8 Matrix Processor in SystemVerilog as an AXI peripheral. Created a Microblaze system to communicate with the Matrix Processor and compute neural network/matrix data.
- 1.0:** Developed a fp 16 matrix processor to communicate with a RISC co-processor using a VLIW scheme. Created test-benches to simulate and verify each module of the design.

Miscellaneous

RTL and TB Generator Scripts
RISC Processor design/verification
HDMI to Ethernet System

Custom Built Electric Guitars
Digital Real Time Echo Generator
Pong redesigned in Verilog

M6800 8-Bit Computer with I/OP
Guitar Hero with 74LS family
Multiplexed AM Telephone Lines

SKILLS

Languages Verilog, SystemVerilog, C, Perl, Python, Tcl, L^AT_EX, Various assembly languages

Tools Vivado & Vitis, Synplify, Virtuosio, Quartus, Git, Modelsim, MATLAB

Hardware Xilinx FPGAs, Altera FPGAs, MSP430, Lab Testing Equipment